

AI VOICE ASSISTANT & SYSTEM AUTOMATION

Project Overview

AI-powered desktop voice assistant automating computer tasks with speech recognition, Google Gemini API intelligence, and MediaPipe gesture controls.

Problem Statement:	Executing repetitive desktop tasks requires manual mouse and keyboard navigation.
Solution Implemented:	Built a Python desktop voice assistant leveraging Gemini API intent parsing, speech synthesis, and MediaPipe vision controls.
My Technical Role:	Lead AI & Voice Developer

Main Features & Technical Stack

Core Features:	Technologies Used:
- Real-time speech recognition & text-to-speech voice synthesis - Google Gemini API integration for intelligent intent parsing - MediaPipe & OpenCV gesture recognition for touchless controls - PowerShell system automation scripts for launching desktop apps	Python, Google Gemini API, SpeechRecognition, OpenCV, MediaPipe, PowerShell
	Architecture Flow:
	Mic / Camera Input → Speech Engine → Gemini API → PowerShell Exec → Voice Output

Technical Challenge & Results

Key Engineering Challenge: Calibrated dynamic ambient noise thresholds to prevent unwanted background command execution.
Validated Results: Delivered a hands-free desktop voice assistant executing system commands effortlessly.

Live Demo: <https://github.com/prashant-singh-78/jarvis.git> | **GitHub:** <https://github.com/prashant-singh-78/jarvis.git> | **Portfolio:** prashant08.vercel.app